

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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1. Experiment Title:

9.1 Theme Preferences — a web page using HTML, CSS and JavaScript that saves the user's selected light/dark theme in localStorage and sessionStorage, restores it automatically on page load, and allows the preference to be cleared.

9.2 Seminar Schedule Planner — a web page using HTML, CSS and JavaScript with a rowspan/colspan seminar timetable, where clicking any seminar topic displays an alert box with the schedule details and saves the selection to localStorage, with options to reload or clear the saved data and a persistent light/dark theme.

2. Software/Tools Required:

Google Chrome / Safari (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, JavaScript, Web Storage API (localStorage & sessionStorage)

3. Experiment Program Code:

9.1 — Theme Preferences

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Theme Preferences</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      padding: 40px;
      background-color: #ffffff;
      color: #000000;
    }

    body.dark {
      background-color: #222222;
      color: #ffffff;
    }

    h1 { margin-bottom: 20px; }

    button {
      padding: 10px 20px;
      margin: 8px;
      font-size: 16px;
      cursor: pointer;
    }
  </style>
</head>
<body>

  <h1>Theme Preferences</h1>
  <p>Using localStorage and sessionStorage to save user preferences.</p>

  <div>
```

```

<button onclick="setTheme('light')">Light Theme</button>
<button onclick="setTheme('dark')">Dark Theme</button>
<button onclick="clearPreference()">Clear Preference</button>
</div>

<script>
// Apply theme and save to both storages
function setTheme(theme) {
  if (theme === 'dark') {
    document.body.classList.add('dark');
  } else {
    document.body.classList.remove('dark');
  }

  // Save to localStorage (persists even after browser is closed)
  localStorage.setItem('theme', theme);

  // Save to sessionStorage (clears when tab/browser is closed)
  sessionStorage.setItem('sessionTheme', theme);

}

// Clear preference from both storages
function clearPreference() {
  localStorage.removeItem('theme');
  sessionStorage.removeItem('sessionTheme');
  document.body.classList.remove('dark');
}

// On page load, restore saved theme from localStorage
window.onload = function () {
  var saved = localStorage.getItem('theme');
  if (saved === 'dark') {
    document.body.classList.add('dark');
  }
};
</script>
</body>
</html>

```

9.2 — Seminar Schedule Planner

[index.html](#)

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Practical 9: Schedule Planner & Local Storage</title>
  <link rel="stylesheet" href="style.css" />
</head>
<body>

  <!-- Theme Toggle Bar -->
  <div class="top-bar">
    <div class="theme-controls">
      <span>Theme: </span>
      <button class="theme-btn" onclick="setTheme('light')">* Light</button>
      <button class="theme-btn" onclick="setTheme('dark')">🌙 Dark</button>
      <button class="theme-btn clear-theme-btn" onclick="clearThemePreference()">Reset
Theme</button>
    </div>

```

```

</div>

<div class="main-container">
  <header class="header">
    <h1>Seminar Schedule Planner</h1>
    <p class="subtitle">Click on any seminar topic or schedule to show the Alert Box and save
preference in Local Storage.</p>
  </header>

  <!-- Schedule Table Section -->
  <div class="table-wrapper">
    <table class="schedule-table" id="scheduleTable">
      <thead>
        <tr>
          <th rowspan="3" class="th-main">Day</th>
          <th colspan="3" class="th-main">Seminar</th>
        </tr>
        <tr>
          <th colspan="2" class="th-sub">Schedule</th>
          <th rowspan="2" class="th-sub">Topic</th>
        </tr>
        <tr>
          <th class="th-time">Begin</th>
          <th class="th-time">End</th>
        </tr>
      </thead>
      <tbody>
        <!-- Monday -->
        <tr>
          <td rowspan="2" class="day-cell">Monday</td>
          <td rowspan="2" class="time-cell bg-yellow">8:00 a.m.</td>
          <td rowspan="2" class="time-cell bg-purple">5:00 p.m.</td>
          <td class="topic-cell" data-day="Monday" data-begin="8:00 a.m." data-end="5:00 p.m."
data-topic="Introduction to XML">
            Introduction to XML
          </td>
        </tr>
        <tr>
          <td class="topic-cell" data-day="Monday" data-begin="8:00 a.m." data-end="5:00 p.m."
data-topic="Validity: DTD and Relax NG">
            Validity: DTD and Relax NG
          </td>
        </tr>

        <!-- Tuesday -->
        <tr>
          <td rowspan="3" class="day-cell">Tuesday</td>
          <td class="time-cell bg-yellow">8:00 a.m.</td>
          <td class="time-cell bg-yellow">11:00 a.m.</td>
          <td rowspan="2" class="topic-cell" data-day="Tuesday" data-begin="8:00 a.m. - 2:00
p.m." data-end="2:00 p.m." data-topic="XPath">
            XPath
          </td>
        </tr>
        <tr>
          <td class="time-cell bg-green">11:00 a.m.</td>
          <td class="time-cell bg-green">2:00 p.m.</td>
        </tr>
        <tr>
          <td class="time-cell bg-green">2:00 p.m.</td>
          <td class="time-cell bg-purple">5:00 p.m.</td>
          <td class="topic-cell" data-day="Tuesday" data-begin="2:00 p.m." data-end="5:00 p.m."
data-topic="XSL Transformations">
            XSL Transformations
          </td>
        </tr>

        <!-- Wednesday -->
        <tr>

```

```

        <td class="day-cell">Wednesday</td>
        <td class="time-cell bg-yellow">8:00 a.m.</td>
        <td class="time-cell bg-green">12:00 p.m.</td>
        <td class="topic-cell" data-day="Wednesday" data-begin="8:00 a.m." data-end="12:00
p.m." data-topic="XSL Formatting Objects">
            XSL Formatting Objects
        </td>
    </tr>
</tbody>
</table>
</div>

<!-- Local Storage Display Section -->
<div class="storage-card">
    <div class="card-header">
        <h2>📁 Saved in Local Storage</h2>
        <span id="storageBadge" class="badge">No Preference Saved</span>
    </div>

    <div id="savedDetails" class="saved-details empty-state">
        <p>No seminar selected yet. Click any seminar above to trigger an alert and save it to
Local Storage.</p>
    </div>

    <div class="action-buttons">
        <button id="clearStorageBtn" class="btn btn-danger" onclick="clearSavedSchedule()">Clear
Saved Schedule</button>
        <button class="btn btn-secondary" onclick="reloadFromStorage()">Reload from
Storage</button>
    </div>
</div>

<script src="script.js"></script>
</body>
</html>

```

style.css

```

/* Reset & Base Styles */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", Arial,
sans-serif;
    background-color: #f4f6f9;
    color: #212529;
    line-height: 1.6;
    padding-bottom: 40px;
    transition: background-color 0.3s ease, color 0.3s ease;
}

/* Dark Theme */
body.dark {
    background-color: #1a1d20;
    color: #f8f9fa;
}

body.dark .main-container {
    background-color: transparent;
}

body.dark .table-wrapper,
body.dark .storage-card {
    background-color: #24282c;
}

```

```
    box-shadow: 0 4px 16px rgba(0, 0, 0, 0.4);
}

body.dark .schedule-table th,
body.dark .schedule-table td {
    border-color: #495057;
    color: #212529;
    /* keep cell text readable with pastel colors */
}

body.dark .schedule-table .th-main,
body.dark .schedule-table .th-sub,
body.dark .schedule-table .th-time,
body.dark .schedule-table .day-cell {
    background-color: #343a40;
    color: #f8f9fa;
}

body.dark .schedule-table .topic-cell {
    background-color: #2b3035;
    color: #f8f9fa;
}

body.dark .schedule-table .topic-cell:hover {
    background-color: #3d444b;
}

body.dark .storage-card h2 {
    color: #f8f9fa;
}

/* Top Theme Bar */
.top-bar {
    display: flex;
    justify-content: flex-end;
    padding: 12px 24px;
    background-color: rgba(255, 255, 255, 0.8);
    backdrop-filter: blur(8px);
    border-bottom: 1px solid #e2e8f0;
}

body.dark .top-bar {
    background-color: rgba(36, 40, 44, 0.8);
    border-bottom-color: #343a40;
}

.theme-controls {
    display: flex;
    align-items: center;
    gap: 8px;
    font-size: 14px;
    font-weight: 500;
}

.theme-btn {
    padding: 6px 12px;
    font-size: 13px;
    border-radius: 6px;
    border: 1px solid #cbd5e1;
    background-color: #ffffff;
    color: #334155;
    cursor: pointer;
    transition: all 0.2s ease;
}

.theme-btn:hover {
    background-color: #f1f5f9;
    border-color: #94a3b8;
}
```

```

body.dark .theme-btn {
  background-color: #2b3035;
  color: #e2e8f0;
  border-color: #4b5563;
}

body.dark .theme-btn:hover {
  background-color: #374151;
}

/* Main Container */
.main-container {
  max-width: 860px;
  margin: 30px auto;
  padding: 0 20px;
}

/* Header */
.header {
  text-align: center;
  margin-bottom: 28px;
}

.header h1 {
  font-size: 26px;
  font-weight: 700;
  color: #1e293b;
  margin-bottom: 8px;
}

body.dark .header h1 {
  color: #f1f5f9;
}

.header .subtitle {
  font-size: 15px;
  color: #64748b;
}

body.dark .header .subtitle {
  color: #94a3b8;
}

/* Table Wrapper */
.table-wrapper {
  background-color: #ffffff;
  padding: 24px;
  border-radius: 12px;
  box-shadow: 0 4px 12px rgba(0, 0, 0, 0.06);
  margin-bottom: 28px;
  overflow-x: auto;
}

/* Schedule Table */
.schedule-table {
  width: 100%;
  border-collapse: collapse;
  text-align: center;
  border: 2px solid #2d3748;
}

.schedule-table th,
.schedule-table td {
  border: 1.5px solid #2d3748;
  padding: 12px 14px;
  font-size: 15px;
  vertical-align: middle;
}

```

```

/* Table Header */
.schedule-table th {
  background-color: #e2e8f0;
  font-weight: 700;
  color: #1e293b;
}

.schedule-table .th-main {
  font-size: 16px;
  padding: 10px;
}

.schedule-table .th-sub {
  font-size: 15px;
  padding: 8px;
}

.schedule-table .th-time {
  font-size: 14px;
  padding: 6px;
}

/* Table Body */
.schedule-table .day-cell {
  font-weight: 600;
  background-color: #f8fafc;
  color: #1e293b;
  width: 15%;
}

/* Cell Colors from Reference Table */
.bg-yellow {
  background-color: #e8ec98 !important;
  /* Soft Pastel Yellow */
  color: #212529 !important;
  font-weight: 500;
}

.bg-purple {
  background-color: #a4a9f3 !important;
  /* Soft Pastel Purple/Blue */
  color: #212529 !important;
  font-weight: 500;
}

.bg-green {
  background-color: #b9edba !important;
  /* Soft Pastel Light Green */
  color: #212529 !important;
  font-weight: 500;
}

/* Interactive Topic Cells */
.topic-cell {
  background-color: #ffffff;
  color: #1e293b;
  cursor: pointer;
  text-align: center;
  font-weight: 500;
  transition: all 0.2s ease;
  position: relative;
}

.topic-cell:hover {
  background-color: #eff6ff !important;
  color: #1d4ed8 !important;
  font-weight: 600;
  transform: scale(1.01);
}

```

```

    box-shadow: inset 0 0 0 2px #3b82f6;
}

.topic-cell.active-selected {
    outline: 3px solid #2563eb;
    background-color: #dbeafe !important;
    color: #1e40af !important;
    font-weight: 700;
}

/* Storage Card */
.storage-card {
    background-color: #ffffff;
    padding: 24px;
    border-radius: 12px;
    box-shadow: 0 4px 12px rgba(0, 0, 0, 0.06);
}

.card-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    margin-bottom: 16px;
    border-bottom: 1px solid #e2e8f0;
    padding-bottom: 12px;
}

body.dark .card-header {
    border-bottom-color: #343a40;
}

.card-header h2 {
    font-size: 18px;
    font-weight: 600;
    color: #1e293b;
}

.badge {
    font-size: 12px;
    font-weight: 600;
    padding: 4px 10px;
    border-radius: 20px;
    background-color: #e2e8f0;
    color: #64748b;
}

.badge.saved {
    background-color: #dcfce7;
    color: #166534;
}

/* Saved Details */
.saved-details {
    background-color: #f8fafc;
    border: 1px solid #e2e8f0;
    border-radius: 8px;
    padding: 16px;
    margin-bottom: 20px;
}

body.dark .saved-details {
    background-color: #1e2227;
    border-color: #374151;
}

.saved-details.empty-state p {
    color: #94a3b8;
    font-style: italic;
    text-align: center;
}

```

```

}

.saved-info-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 12px;
}

.saved-info-item {
  display: flex;
  flex-direction: column;
}

.saved-info-item .info-label {
  font-size: 12px;
  font-weight: 600;
  color: #64748b;
  text-transform: uppercase;
}

.saved-info-item .info-value {
  font-size: 15px;
  font-weight: 600;
  color: #0f172a;
}

body.dark .saved-info-item .info-value {
  color: #f1f5f9;
}

/* Action Buttons */
.action-buttons {
  display: flex;
  gap: 12px;
  justify-content: flex-end;
}

.btn {
  padding: 9px 18px;
  font-size: 14px;
  font-weight: 500;
  border-radius: 6px;
  cursor: pointer;
  border: none;
  transition: all 0.2s ease;
}

.btn-danger {
  background-color: #ef4444;
  color: #ffffff;
}

.btn-danger:hover {
  background-color: #dc2626;
}

.btn-secondary {
  background-color: #64748b;
  color: #ffffff;
}

.btn-secondary:hover {
  background-color: #475569;
}

/* Responsive */
@media (max-width: 650px) {
  .schedule-table th,

```

```

.schedule-table td {
  padding: 8px 6px;
  font-size: 13px;
}

.saved-info-grid {
  grid-template-columns: 1fr;
}
}

```

script.js

```

// =====
// Seminar Schedule Planner & Local Storage
// =====

// Key used for saving seminar schedule in localStorage
const STORAGE_KEY = 'selectedSeminar';
const THEME_KEY = 'themePreference';

// Attach event listeners when DOM content is loaded
document.addEventListener('DOMContentLoaded', () => {
  initTheme();
  initTableClickEvents();
  renderSavedSchedule();
});

/**
 * Attaches Click Event Listeners to all Topic cells in the table
 */
function initTableClickEvents() {
  const topicCells = document.querySelectorAll('.topic-cell');

  topicCells.forEach(cell => {
    cell.addEventListener('click', () => {
      const day = cell.getAttribute('data-day');
      const begin = cell.getAttribute('data-begin');
      const end = cell.getAttribute('data-end');
      const topic = cell.getAttribute('data-topic');

      // 1. Show the Alert Box (as required)
      alert(
        `
        Seminar Schedule Details:\n` +
        `-----\n` +
        `Day      : ${day}\n` +
        `Time      : ${begin} to ${end}\n` +
        `Topic      : ${topic}\n` +
        `-----\n` +
        `✅ Saved to Local Storage!`
      );

      // 2. Save Selection to Local Storage
      const scheduleData = {
        day: day,
        begin: begin,
        end: end,
        topic: topic,
        savedAt: new Date().toLocaleString()
      };

      localStorage.setItem(STORAGE_KEY, JSON.stringify(scheduleData));

      // 3. Update the UI & highlight active cell
      highlightSelectedCell(topic);
      renderSavedSchedule();
    });
  });
}

/**

```

```

    * Reads from Local Storage and renders the saved schedule details
    */
function renderSavedSchedule() {
    const savedDataRaw = localStorage.getItem(STORAGE_KEY);
    const savedDetailsContainer = document.getElementById('savedDetails');
    const storageBadge = document.getElementById('storageBadge');

    if (!savedDataRaw) {
        savedDetailsContainer.className = 'saved-details empty-state';
        savedDetailsContainer.innerHTML = `
            <p>No seminar selected yet. Click any seminar above to trigger an alert and save it to
Local Storage.</p>
        `;
        storageBadge.className = 'badge';
        storageBadge.textContent = 'No Preference Saved';
        clearHighlight();
        return;
    }

    try {
        const data = JSON.parse(savedDataRaw);

        savedDetailsContainer.className = 'saved-details';
        savedDetailsContainer.innerHTML = `
            <div class="saved-info-grid">
                <div class="saved-info-item">
                    <span class="info-label">Day:</span>
                    <span class="info-value">${data.day}</span>
                </div>
                <div class="saved-info-item">
                    <span class="info-label">Time Schedule:</span>
                    <span class="info-value">${data.begin} - ${data.end}</span>
                </div>
                <div class="saved-info-item" style="grid-column: span 2;">
                    <span class="info-label">Topic:</span>
                    <span class="info-value">${data.topic}</span>
                </div>
                <div class="saved-info-item" style="grid-column: span 2;">
                    <span class="info-label">Saved At:</span>
                    <span class="info-value" style="font-size: 13px; color:
#64748b;">${data.savedAt}</span>
                </div>
            </div>
        `;

        storageBadge.className = 'badge saved';
        storageBadge.textContent = '● Saved in LocalStorage';

        highlightSelectedCell(data.topic);
    } catch (e) {
        console.error('Error parsing stored schedule data:', e);
    }
}

/**
 * Highlights the active cell matching the saved topic
 */
function highlightSelectedCell(topicName) {
    clearHighlight();
    const topicCells = document.querySelectorAll('.topic-cell');
    topicCells.forEach(cell => {
        if (cell.getAttribute('data-topic') === topicName) {
            cell.classList.add('active-selected');
        }
    });
}

/**
 * Clears active highlight class from all cells

```

```

*/
function clearHighlight() {
  const topicCells = document.querySelectorAll('.topic-cell');
  topicCells.forEach(cell => {
    cell.classList.remove('active-selected');
  });
}

/**
 * Clears the saved schedule from Local Storage
 */
function clearSavedSchedule() {
  const existing = localStorage.getItem(STORAGE_KEY);
  if (existing) {
    localStorage.removeItem(STORAGE_KEY);
    renderSavedSchedule();
    alert('🗑️ Saved seminar schedule removed from Local Storage.');
  } else {
    alert('No schedule is currently saved in Local Storage.');
  }
}

/**
 * Reloads the stored data from Local Storage manually
 */
function reloadFromStorage() {
  renderSavedSchedule();
  alert('🔄 Checked and reloaded latest state from Local Storage.');
}

// =====
// Theme Management (Light / Dark with localStorage)
// =====

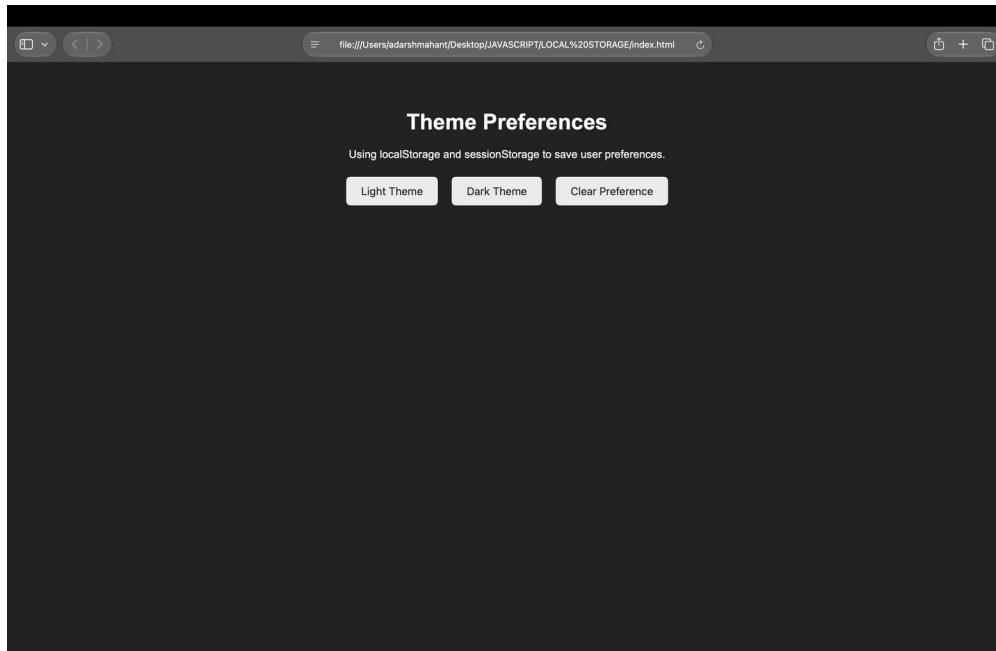
function setTheme(theme) {
  if (theme === 'dark') {
    document.body.classList.add('dark');
  } else {
    document.body.classList.remove('dark');
  }
  localStorage.setItem(THEME_KEY, theme);
  sessionStorage.setItem('sessionTheme', theme);
}

function clearThemePreference() {
  localStorage.removeItem(THEME_KEY);
  sessionStorage.removeItem('sessionTheme');
  document.body.classList.remove('dark');
  alert('Theme preference reset to default.');
```

4. Output:

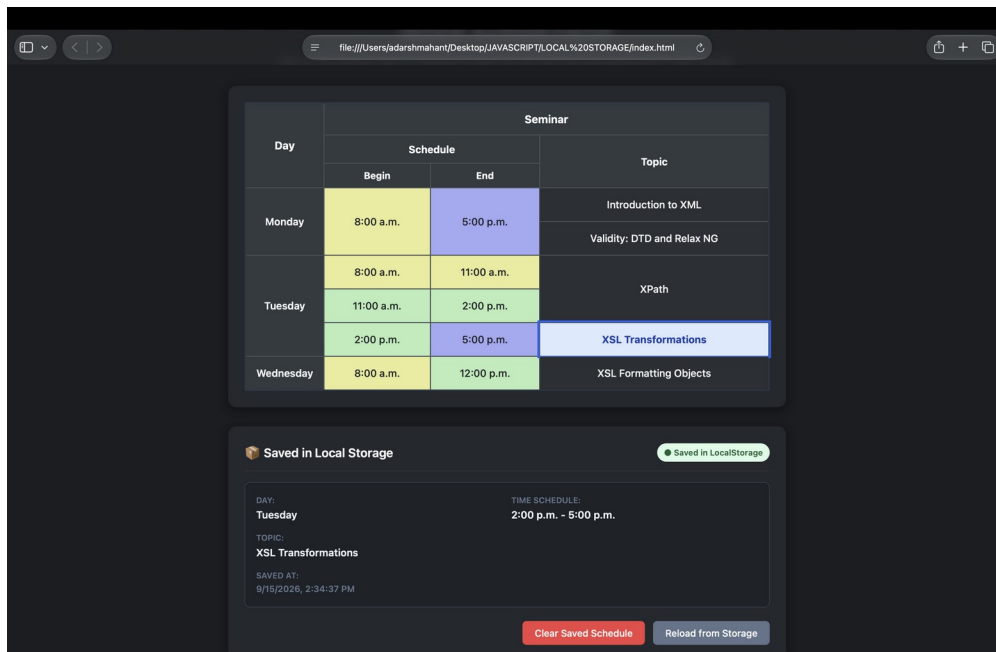
9.1 — Theme Preferences

PRN: 24070521038 | Name: Vansh Mahant | File: /JAVASCRIPT/LOCAL STORAGE/index.html — Dark theme applied and saved in storage



9.2 — Seminar Schedule Planner

PRN: 24070521038 | Name: Vansh Mahant | File: /JAVASCRIPT/LOCAL STORAGE/index.html — Seminar selected and saved in Local Storage



5. Result/Conclusion:

Two web pages demonstrating browser storage were successfully implemented using HTML, CSS and JavaScript. Part 9.1 stores the user's chosen light or dark theme in both localStorage and sessionStorage, restores it automatically using the window.onload event, and removes it when the preference is cleared. Part 9.2 builds a seminar timetable using rowspan and colspan, attaches click event listeners to each topic cell so that an alert box displays the day, time schedule and topic, and stores the selection as a JSON object in localStorage with a timestamp — which is then read back and rendered in a summary card, with the matching cell highlighted, plus buttons to

reload or clear the stored data. This experiment reinforced the use of the Web Storage API (localStorage and sessionStorage), JSON serialization, event handling, and persisting user preferences across sessions.

Upload the Soft copy as well as complete code on GitHub link is mandatory.

File name: Practical 9_24070521038_Vansh Mahant_A_A1